

Marcos Rodrigo Talavera

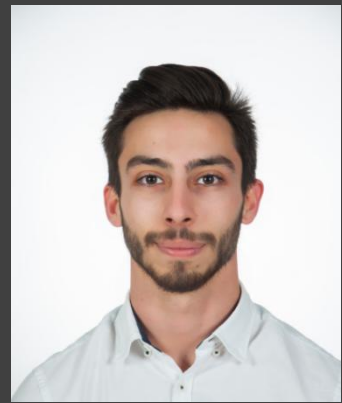
Telecommunications Engineer

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Education

PhD in Communication Technologies and Systems (2020-2025)

Universidad Politécnica de Madrid, UPM

Master's Degree in Telecommunications Engineering (2018-2020)

Universidad Politécnica de Madrid, UPM

- Master's Thesis grade: Distinction (10/10)

Bachelor's Degree in Telecommunications Systems Engineering (2013-2018)

Universidad de Alcalá de Henares, UAH

- Bachelor's Thesis grade: Excellent (9/10)

Supplementary Education

How Diffusion Models Work (2025)

DeepLearning.AI

Generative AI with Large Language Models (2024)

DeepLearning.AI, Coursera, AWS

Deep Learning Specialization (2023)

DeepLearning.AI, Coursera

English as a Second Language (2015-2016)

University of California Los Angeles, UCLA

Campus Party Technology Event, England (2013)

Universidad de Alcalá de Henares, UAH

Program in Cognitive Development Techniques (2010-2012)

SAPIENTEC

Mathematical Thinking School (2006-2009)

Escuela de Pensamiento Matemático Miguel de Guzmán

Professional Experience

E.T.S.I. Telecomunicación, Universidad Politécnica de Madrid (2020-present)

Post-graduate researcher in the Grupo de Tratamiento de Imágenes, Departamento de Señales, Sistemas y Radiocomunicaciones

Design and implementation of deep-learning techniques for computer vision, covering the detection and identification of people with convolutional neural networks and Vision Transformers, as well as long-form video event recognition through 2D/3D CNNs, recurrent networks and Transformers.

Development of specific strategies for vehicle detection and tracking, creation of ad-hoc databases for training AI algorithms, and system-administration tasks (user management, package control, drivers, etc.). These implementations were carried out in close R&D collaborations with NOKIA and AIRBUS laboratories on various projects.

Active collaboration in teaching within the *Grado en Ingeniería de Tecnologías y Servicios de Telecomunicación* (GITST), *Grado en Ingeniería y Sistemas de Datos* (GISD) and *Máster Universitario en Ingeniería de Telecomunicación* (MUIT).

Supervision of two Master's theses focused respectively on subject identification with neural networks and on automatic soccer-match summarisation using multimodal (text + image) neural networks.

Urban Planet (2018)

Sports instructor

Sports instructor at this leisure centre while completing my Master's studies, responsible for teaching classes, accident prevention and leading group activities. I also provided direct customer service, mainly assisting English-speaking visitors.

Indra Sistemas S.A. (2017-2018)

Junior programmer in the Air Traffic Management Department

Development and implementation of scripts for Voice-over-IP communication systems in air-traffic-management environments. Task automation, handling of telecommunications equipment, and configuration and connection of networks, routers and switches under Scrum methodology.

DEYDE Calidad de Datos S.L. (2016-2017)

Data coder and trainee programmer for expert systems in postal-address coding

Updating and coding master files of municipalities and streets, collaborating in the technical development of expert systems for postal-address correction and coding.

Self-employed (2013-2018)

Private tutor of Mathematics, Physics, Chemistry and English for 1st and 2nd year *Bachillerato* students

Alongside my university studies, I delivered private lessons in mathematics, physics, chemistry and English to final-year secondary students, focusing on those preparing for university-entrance exams.

Research Activity (Publications)

Text-Guided Sports Highlights: A CLIP-Based Framework for Automatic Video Summarization (under review, 2025)

Rodrigo, M., Cuevas, C. & García, N. Text-guided sports highlights: a CLIP-based framework for automatic video summarization. *IEEE Transactions on Consumer Electronics* (under review).

ViMoCLIP: Augmenting Static CLIP Representations with Video Motion Cues for Animal Action Recognition (2025)

Rodrigo, M., Zhong, E., del Blanco, C. R., Cuevas, C., Jaureguizar, F. & García, N. ViMoCLIP: augmenting static CLIP representations with video-motion cues for animal-action recognition. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops (2025).

Comprehensive Comparison Between Vision Transformers and Convolutional Neural Networks for Face Recognition Tasks (2024)

Rodrigo, M., Cuevas, C. & García, N. Comprehensive comparison between vision transformers and convolutional neural networks for face recognition tasks. *Scientific Reports* 14, 21392 (2024). <https://doi.org/10.1038/s41598-024-72254-w>.

Automatic Highlight Detection in Videos of Martial Arts Tricking (2023)

M. Rodrigo, C. Cuevas, D. Berjón, and N. García. Automatic highlight detection in videos of martial arts tricking. *Multimedia Tools Applications*, 2023, doi: 10.1007/s11042-023-16003-7

UPM-GTI-Face: A Dataset for the Evaluation of the Impact of Distance and Masks in Face Detection and Recognition Systems (2022)

M. Rodrigo, E. González-Sosa, C. Cuevas, and N. García. UPM-GTI-Face: A dataset for the evaluation of the impact of distance and masks in face detection and recognition systems. *IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)*, Madrid, Spain, 2022, pp. 1-8, doi: 10.1109/AVSS56176.2022.9959558.

Development of a Strategy to Automatically Detect Highlights in Martial Arts Tricking Videos (2020)

M. Rodrigo. Development of a strategy to automatically detect highlights in martial arts tricking videos. Thesis (Master thesis), E.T.S.I. Telecomunicación (UPM).

Acoustic Classification of Birds by Extracting Sound Features Using MFCC Parameters (2018)

M. Rodrigo. Clasificación acústica de aves mediante la extracción de características del sonido a través de los parámetros MFCC. Thesis (Bachelor thesis), E.P.S. (UAH).

Languages

Spanish – Native

English – Bilingual

- Certified C1 level (University of Cambridge)
- University of California Los Angeles, UCLA (1 year)
- Mentor Academy of Beverly Hills (1 month)
- British Council (2 years)

French – Lower-intermediate level

- Gredos San Diego School (6 years)

Technical Skills

Operating Systems

Microsoft Windows, Linux.

Neural Networks

MLPs, CNNs, RNNs, LSTMs, GRUs, Siamese Networks, Distilled Networks, Autoencoders, VAEs, GANs, Diffusion Models, Transformers, Vision Transformers, LLMs, VLMs, MoEs.

Document and Presentation Authoring

Microsoft Word, PowerPoint, Excel, Prezi, Photoshop, Premiere, After Effects.

Programming Languages

Python, C, C++, C#, HTML, CSS, JavaScript, SQL, MATLAB, Bash.

Tools, Programs and Libraries

TensorFlow, Keras, PyTorch, HuggingFace, OpenCV, Pandas, MySQL, Kubernetes, Docker, Streamlit, Gradio.

Personal Skills

Self-taught. Resourceful. Proactive. Teamwork. Leadership capacity. Technology communicator. Teaching enthusiast. Knowledge-sharing facilitator.

Personal Projects

Implementation of a Kubernetes Cluster for Teaching and Research Environments

I developed, together with a colleague, a Kubernetes cluster that integrates 17 GPU-equipped computers to optimise resources for academic and research tasks. We configured support for NVIDIA cards, native authentication and user management, with a single HTTPS-based public access point.

We also set up an NFS storage system, custom profiles with Docker images and multiple Python versions. To simplify usage, we created a minimalist JupyterHub interface that lets users assign nodes and containers quickly, enabling distributed neural-network training.

Technologies used: Docker, Kubernetes, JupyterHub, NFS, SSH, HTTPS, TensorFlow, PyTorch and Python.

RoboMaster Tank Detection with Computer Vision

During my third PhD year, I designed from scratch a practical assignment that represented one third of a fourth-year course in the *Grado en Ingeniería y Sistemas de Datos*. We purchased several RoboMaster tanks (small robot vehicles with cannon, sensors and cameras) and integrated them with a Python SDK for control.

I developed three main computer-vision approaches: a localisation network built from scratch in TensorFlow (manually defining the model and loss functions); an object-detection network in PyTorch for identifying multiple targets in an image; and finally, the implementation of YOLOv8 for real-time detection.

I also lectured on common object-detection methods and their evolution and supervised students so they could move their robots autonomously and detect rivals in the game arena.

Technologies used: Python, TensorFlow, PyTorch, YOLO and SDK.

Breakfast-Order Management Web App

In my spare time I built a web application to organise and optimise breakfast orders for my 20-person research group. The system automatically calculates the most cost-effective drink-and-food combinations and incorporates a debt-tracking module that replaces the third-party app we previously used.

I implemented the solution in Python, using Streamlit for the user interface, and deployed it on a Linux server behind a reverse proxy that routes requests from the group's website. The app runs continuously, simplifying order taking and shared-expense tracking.

Technologies used: Python, Streamlit, Linux, Reverse Proxy, HTTPS and SSH.

Story and Image Generation with Generative Models (Llama, Stable Diffusion)

I developed a system that, integrated with the breakfast-ordering app, automatically creates fictional stories and portraits of my research-group members. I used Llama 3.3 to craft narratives describing how each colleague becomes a supposed "defaulter," while Stable Diffusion 3.5 Large generates the accompanying images.

What began as a humorous project has become an essential add-on: every morning a Linux service processes a JSON file with each person's info and produces new stories and images, adding a creative, fun touch to breakfast time.

Technologies used: Python, Llama 3.3, Stable Diffusion 3.5 Large, Linux and JSON.

Selecting Components and Assembling Computers for Deep-Learning Development

Several years ago, I built my first desktop PC, investing many hours in component selection and compatibility checks. Since then, I have assembled three more personal systems and others for friends and family, optimising price-to-performance across configurations.

One standout build is a machine tailored to run neural networks, powered by a high-end NVIDIA GPU and specs focused on AI development. This experience led me to oversee component selection and assembly of multiple high-performance, multi-GPU workstations for my research group, which we use daily.

Beyond hardware integration, I handled system-administration tasks: installing and configuring various OSs, managing users, controlling package versions and keeping drivers up to date.

Technologies used: GPU architectures (NVIDIA), PC hardware (motherboards, CPUs, RAM, SSDs), Linux, System Administration, Package Management, Drivers.

INDESIHack: Adverse-Weather Detection for Ferrovial

I took part in this hackathon aimed at identifying visibility issues on roads from traffic-camera images. I led a team comprising a PhD colleague and two participants we met during the challenge.

We built a “model of experts” (MoE) that combined several tools—including CLIP (a vision-language model), Microsoft Azure and the ChatGPT API—to assess visibility quality in images captured worldwide. The project ran on AWS SageMaker, enabling scalability and large-scale data processing.

Technologies used: CLIP (VLM), Microsoft Azure, AWS SageMaker, ChatGPT API, Python and MoE.

Custom-Components Workshop with TensorFlow

After studying *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, I organised a series of workshops to share what I had learned. These sessions, initially for colleagues, soon welcomed members of the wider university community.

During the meetings I demonstrated how to create custom TensorFlow components (loss functions, activations, initialisers, regularisers, metrics, layers, models and training loops) and we explored the library's internals, including graph management and model tracing.

Technologies used: Python, TensorFlow, Keras.

Personal Website Development

To understand the process of building websites from scratch, I taught myself HTML and CSS through tutorials and, starting from a simple template, created my own site.

I learned to structure content with HTML and style it with CSS, incorporating elements and sections that reflect my professional path and interests. The page is publicly hosted on GitHub Pages, which familiarised me with Git for version control and cloud repositories.

This project gave me practical insight into designing, deploying and maintaining simple yet functional websites.

Technologies used: HTML, CSS, JavaScript, Git and GitHub Pages.

Reading Club

I started a reading club with workmates to delve into AI-related technical books. Every one or two weeks, depending on text complexity, we choose one or more chapters and meet to discuss findings, exchange views and analyse model-implementation details. These sessions keep us up to date, promote collaborative learning and strengthen our skills in the AI field

Courses and Talks

Throughout my university career I encouraged the organisation of regular meetings to discuss recent technological advances, scientific papers and practical tools. This initiative evolved into collaborative gatherings where each session is led by a volunteer who explores a chosen topic and shares theoretical or methodological foundations with the group. These talks have fostered continuous learning and a culture of collaboration.

Relevant Reading

Hands-On Machine Learning with Scikit-learn, Keras, and TensorFlow

Aurélien Géron (2022)

Deep Learning with Python

François Chollet (2021)

Hands-On Generative AI with Transformers and Diffusion Models

Omar Sanseviero (2024)

Unix and Linux System Administration Handbook

Evi Nemeth (2017)

Neural Networks from Scratch in Python

Harrison Kinsley (2020)

Machine Learning with PyTorch and Scikit-learn

Sebastian Raschka (2022)

Natural Language Processing with Transformers

Leandro von Werra (2022)

Deep Learning

Ian Goodfellow (2016)

Personal Interests

Passion for New Technologies and AI

I am enthusiastic about keeping up with the latest technological innovations and therefore frequently attend talks, exhibitions and workshops on Artificial Intelligence. I also devote part of my free time to following tutorials and guides—mainly in audiovisual format (YouTube) but also on specialised blogs—which has allowed me to acquire new skills in programming languages such as C++ and C#, as well as in container tools like Docker and Kubernetes.

Thanks to this ongoing learning process, I constantly identify new opportunities to apply AI in different fields and broaden my professional competences.

Image and Video Generation with ComfyUI

The ComfyUI tool has enabled me to immerse myself in the fascinating world of diffusion models for generating images and videos. This experience naturally combines my interest in AI with my creative side, opening up a wide range of possibilities for multimedia content production.

Martial Arts and Acrobatics

I have been practising tricking, gymnastics and acrobatics for more than ten years, and I have also trained in related disciplines such as taekwondo, boxing and kickboxing. This dedication has helped me develop strong discipline, perseverance and concentration. I also occasionally enjoy many other sports—cycling, trials biking, archery, football and parkour—which lets me diversify my skills and maintain an active lifestyle.

Chess

I have recently discovered a passion for this mental sport, raising my ELO from 500 to almost 1300 in just a few months. Several of my close friends have played since childhood, even competing in tournaments, and they were the ones who finally passed on their enthusiasm for chess to me. I am especially motivated by the strategy and continual learning it demands.

Travel and Outdoor Activities

I love exploring the world and immersing myself in different cultures, which has led me to visit more than 25 countries; my favourites are Botswana (where close friends live) and Japan (where I have family ties). In addition to hiking, I enjoy adventure and sporting activities such as bungee jumping and skydiving. These experiences have not only broadened my outlook on the world but have also enhanced my adaptability.